

The Effect of Contamination on Contamination Control Regulation

US Coal Mining and Drinking Water Utilities

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Self-reported compliance

- Self-reported regulation monitoring requires entities to demonstrate compliance to regulators.
- Incentives to shirk reporting:
 - ▶ Concealing non-compliance
 - ▶ Saving self-reporting costs
- Environmental contamination regulation (e.g. US drinking water):
 - ▶ $\uparrow$ contamination $\rightarrow$ $\uparrow$ shirking incentives
- **Why care about self-reporting and contamination?**
 - ▶ Different fix if concealing non-compliance vs self-reporting costs
 - ▶ Monitoring used across industries (HIPAA violation disclosure, SOX, energy firms)
 - ▶ Drinking water for 85% of households is regulated this way

Research question

1. What happens to firm self-reporting when contamination rises?
2. Are changes in self-reporting due to concealment or self-reporting costs?

Context: US drinking water utilities exposed to water contamination from upstream coal mining (1985–2005)

0. Does upstream coal mining raise contamination at drinking water utilities?

How I answer these questions

- Annual panel of drinking water utilities on average 22 km downstream of active coal mines.
- Contaminant concentrations for arsenic, nitrate, barium, and selenium, 1997–2005.
- Utility self-reporting, regulator visits, and enforcement received.
- Instrument upstream coal production.

Why this is appropriate

- Separation between contamination source and regulated firm helps identify the effect of contamination.
- Coal mining → inorganic chemicals (“IOCs”) into surface and ground water.
- Geology of coal seams and national energy market determines mining.

Utility self-reporting nitrate contamination

Utility manager/staff cognitive burden (oleckno1982national).

Self-reporting requirement for nitrates (one of 83 contaminants):

- **Baseline:** once a year, send one sample from every intake to a lab
- **Elevated:** nitrate ≥ 5 mg/L, half the Maximum Contaminant Level ("MCL"), test quarterly

Miss a test, receive a monitoring and reporting ("MR") violation and:

- Avoid an MCL (need self-report for MCL)
- Regulator visit
- Or enforcement: Informal (letters, notices, compliance plans); Formal (consent decrees, fines, prosecution)
- MR punished less severely than MCL

Does coal production raise contamination at drinking water utilities?

$$Concen_{ct} = \beta \cdot \sum_{\tau=1985}^t \text{UpstreamCoalProd}_{c\tau} + \rho_c + \text{Watershed}_c \cdot \tau_t + \epsilon_{ct}$$

- $Concen_{ct}$: mean annual concentration at utility c , year t (SYR2, 1998–2005).
- β : effect of 10 million additional cumulative short tons mined upstream since 1985.
- ρ_c : utility fixed effects. $\text{Watershed}_c \cdot \tau_t$: Watershed-specific year effects, absorbing watershed-wide industrial trends.
- SEs clustered at the utility level.
- Continuous treatment, **no control**: identification comes from relative exposure.
- Assumption: utilities with large upstream production changes would have trended like those with small ones. (balance test on utility characteristics $\rightarrow$)

Mining raises contaminant levels

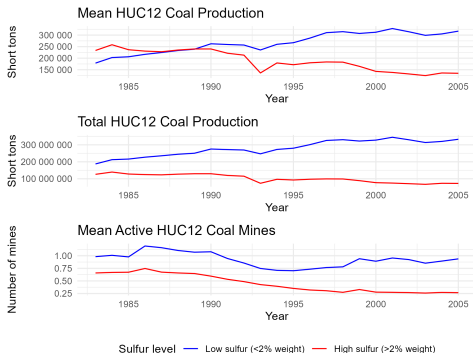
	Arsenic (mg/L) (1)	Nitrate (mg/L) (2)	Barium (mg/L) (3)	Selenium (mg/L) (4)
Cumul. upstream coal prod. (M ST)	0.00001 (0.00001)	0.00883*** (0.00257)	0.00136 (0.00112)	0.00004** (0.00002)
Sample mean (million short tons)	6.9	8.6	6.7	6.7
MCL (mg/L)	0.050	10.000	2.000	0.050
Observations	366	472	353	350

Notes: All specifications include utility and watershed $\times$ year fixed effects. Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

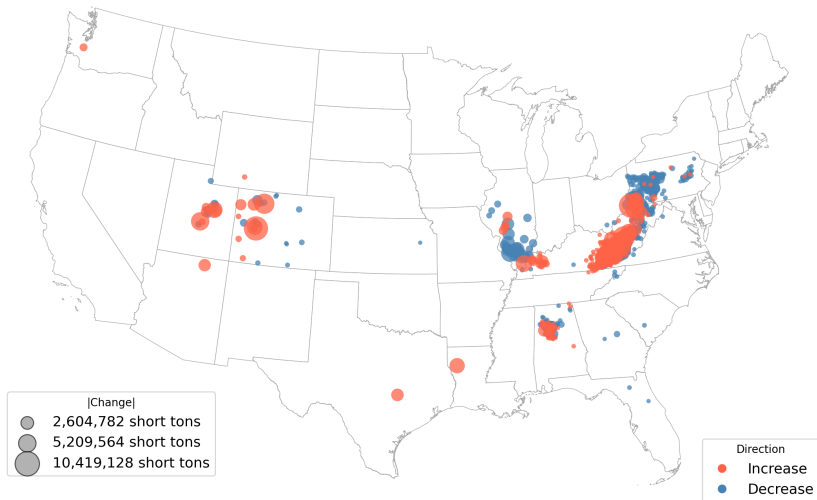
Per sample mean tons mined upstream: nitrate +0.076 mg/L (**), 9.4% of its mean; selenium +0.0003 mg/L (**), 6.9% of its mean. **Every reading in the sample stays under its MCL.** In the sample, only 49 IOC MCL violations.

Coal mining & utility behavior: instrumenting coal mining

- Adapt **stratford2015coalprod** Acid Rain Program (“ARP”) instrument for coal production.
- Use coal sulfur % and the 1995 SO₂ power plant compliance deadline.
- OLS: mining and utility characteristics not random.
- Low sulfur coal was the main compliance response (**lattanzio2022clean**).
- High sulfur mines have limited ability to wash the sulfur out.



Where coal production moved, 1985–2005



IV design: the ARP as a shifter of where coal is mined

First stage:

$$\text{NumMines}_{ct} = \alpha_0 + \alpha_1 \text{NumFacility}_{ct} + \alpha_2 \text{UpstreamCoalSulfur}\%_c \cdot \text{Post95}_t \\ + \text{State}_s \cdot Y_t + C_c + \nu_{ct}$$

Second stage:

$$\text{Outcome}_{vct} = \beta_0 + \beta_1 \text{NumFacility}_{ct} + \gamma \widehat{\text{NumMines}}_{ct} + \text{State}_s \cdot Y_t + C_c + \epsilon_{ct}$$

- $\text{Post95}_t = \mathbf{1}\{t \geq 1995\}$. Sulfur is time-invariant, so C_c absorbs its level.
- Outcome: 1 if utility c had a violation, visit, or enforcement of type v at any point in year t .
- Coefficients are in percentage points.
- SEs clustered at the utility level.

Instrument validity

- **Independence.** The ARP is a national clean-air law set by national energy and emissions politics, not downstream utilities' water quality.
- **Exclusion.** Effect of instrumenting mining downstream of the utility on utility behavior is not significant. (MR violations →), (visits →), (enforcement →).
Utility characteristics are not predicted by the instrument. (full balance table →)
- **Monotonicity.** A defier is a high-sulfur watershed that expanded *because* its coal was high sulfur. Scrubber adoption costs more likely lead to lost rather than gained market share; low-cost high-sulfur regions that expand anyway are never-takers.

Upstream mining reduces utilities' self-reporting

	Monitoring and reporting (MR) violation					
	Nitrates (1)	(2)	Arsenic (3)	(4)	Inorganic chemicals (5)	(6)
OLS						
Upstream coal mines (sum)	0.68** (0.33)	0.55 (0.34)	0.62** (0.31)	0.42 (0.33)	0.49 (0.32)	0.19 (0.34)
2SLS						
Upstream coal mines (sum)	5.12*** (1.70)	3.94** (1.75)	4.85*** (1.57)	3.78** (1.55)	4.66*** (1.62)	3.06* (1.66)
RF						
Post-1995 × Upstream sulfur %	-2.12*** (0.66)	-1.95** (0.83)	-2.01*** (0.60)	-1.87** (0.73)	-1.93*** (0.63)	-1.51* (0.79)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State × year fixed effects		✓		✓		✓

Notes: Coefficients and standard errors are in percentage points. The sample has 565 utilities and 10,641 utility-year observations. Standard errors clustered at the utility level. *** p<0.01, ** p<0.05, * p<0.1.

Base rates of nitrates, arsenic, and IOC are 5.7%, 3.7%, and 4.3%. The 0.56 additional mines a 1-SD lower-sulfur watershed kept after 1995 raise nitrate, arsenic, and IOC MR violations by 2.2, 2.1, and 1.7 pp (state × year fixed effects). OLS estimates are far smaller — consistent with negative omitted variable bias, as expected if poorly performing utilities (more MR violations) are protected from mine siting (fewer mines).

Reporting cost, not concealment

- Nitrate rules require **more frequent monitoring** once a reading exceeds 50% of the MCL — a discrete jump in cost, with no jump in the penalty for under-reporting and no MCL.

	Nitrate MR (1-yr) (1)	Nitrate MR (6-mon) (2)
Concen. > 50% MCL	58.9341** (24.5521)	26.0695** (10.5051)
Mean concen. (z-score)	1.2906 (1.4648)	0.0795 (0.9057)
Observations	835	835

Notes: All specifications include utility and year fixed effects. SEs clustered at the utility level. *** p<0.01, ** p<0.05, * p<0.1.

Descriptive, not causal — read the sign, not the magnitude. Utilities under-report when self-reporting gets expensive, not to conceal exceedances.

The regulator increases inspections

	Sanitary (1)	Technical assistance (2)	Enforcement (3)	Sample collection (4)	Inspection (5)
2SLS					
Upstream coal mines (sum)	0.78 (1.20)	0.67 (0.95)	1.07 (0.81)	0.48 (0.50)	2.13** (1.07)

Notes: Coefficients and standard errors are in percentage points. All specifications include utility and state $\times$ year fixed effects. The sample has 565 utilities and 10,641 utility-year observations. Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Inspection visits rise significantly (+2.1 pp; baseline inspection rate 3.4%). The 0.56 additional mines a 1-SD lower-sulfur watershed kept after 1995 raise inspection visits by 1.2 pp. Regulators do not systematically step up the visit types tied to sanctions. Visits would reveal concealment and be associated with higher penalties.

Enforcement does not counteract under-reporting

	Informal		Enforcement Formal		None	
	(1)	(2)	(3)	(4)	(5)	(6)
2SLS						
Upstream coal mines (sum)	-3.06 (2.15)	-0.71 (2.10)	-3.37*** (0.95)	-1.08 (0.86)	2.37 (2.11)	-0.42 (2.25)
Utility fixed effects	✓	✓	✓	✓	✓	✓
Year fixed effects	✓		✓		✓	
State $\times$ year fixed effects		✓		✓		✓

Notes: Coefficients and standard errors are in percentage points. The sample has 565 utilities and 10,641 utility-year observations. Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Baseline enforcement rates: informal 18.3%, formal 2.9%. One more upstream mine reduces formal enforcement by 3.4 pp with utility and year fixed effects alone (significant; the 0.47 additional mines a 1-SD lower-sulfur watershed kept after 1995 reduce it by 1.6 pp), but the effect loses significance once state $\times$ year fixed effects are added.

Conclusion

- Exogenous contamination raises the cost of self-reporting, and self-reporting falls — **even when utilities remain below MCL**.
- Under-reporting due to cost avoidance, rather than concealment of a contamination violation.
- Regulators' inspection visits rise significantly — substituting for utility self-reporting.
- Reduction in enforcement along with increases in inspection suggests that contaminant exceedances are not being concealed and MR violations are not strongly punished.
- **Policy.** Raising self-reporting penalties or reducing self-reporting costs could reduce under-reporting.
- Open question: whether regulator monitoring is more efficient than self-reporting.

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Appendix

Appendix: instrument balance on intake facilities

	Number of Intake Facilities	
	(1)	(2)
Upstream sulfur		0.9942*** (0.3834)
Post-1995 $\times$ Upstream sulfur	0.0509 (0.0386)	0.0563 (0.0376)
Utility fixed effects	✓	
Year fixed effects		✓
State $\times$ year fixed effects	✓	
Balanced panel		✓
Utilities	565	447
Observations	10,641	9,387

Notes: Standard errors clustered at the utility level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

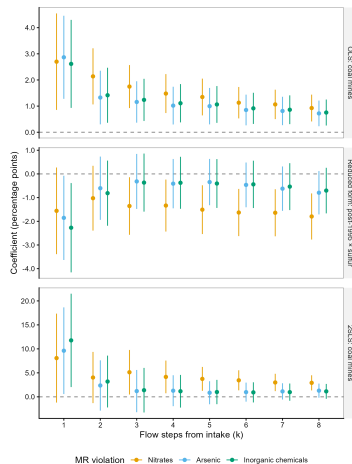
Appendix: balance test on utility characteristics

	Cumul. upstream coal prod. (10M ST)		Any upstream coal mining	
	(1)	(2)	(3)	(4)
Constant	0.2504 (0.2532)		-0.1596 (0.1198)	
Log backcast pop. 1997	0.1091 (0.0750)	0.1512 (0.0938)	0.0159 (0.0126)	0.0208 (0.0144)
Log pop. served (SYR2)	-0.0934 (0.0666)	-0.1113 (0.0956)	0.0060 (0.0259)	-0.0301 (0.0334)
Number of intake facilities	0.0089 (0.0569)	-0.0324 (0.0744)	0.0680* (0.0359)	0.0721* (0.0406)
Surface water source	0.1225 (0.1643)	-0.3902 (0.3296)	0.1210 (0.1307)	0.0521 (0.1601)
Public ownership	-0.1554 (0.2551)	-0.0599 (0.2240)	0.2084** (0.0911)	0.1048 (0.1098)
Wholesaler	-0.3780* (0.2041)	-0.7465 (0.9141)	-0.2164 (0.1340)	-0.2212 (0.1683)
Source water protection	-0.4549 (0.2749)	-0.4385 (0.4305)	-0.2807* (0.1521)	-0.1460 (0.1930)
Joint <i>F</i> -test (all covariates)	2.33	0.96	8.96	2.46
<i>p</i> -value	0.029	0.461	0.000	0.023
Observations	122	120	122	120
State fixed effects		✓		✓

Notes: Heteroskedasticity-robust standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

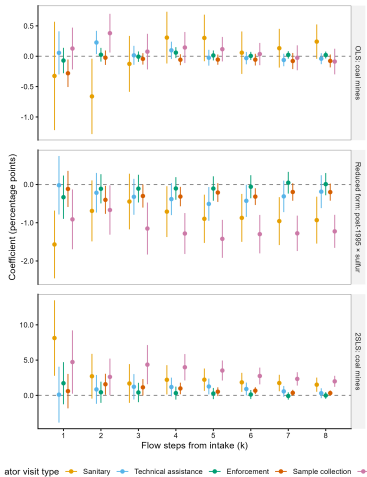
Observable characteristics jointly predict continuous upstream production only without state fixed effects ($p=0.029$) and lose significance once state fixed effects are added ($p=0.461$), as in the main specification. The joint test on the binary presence of upstream mining stays significant with or without state fixed effects.

Appendix: exclusion placebo, MR violations across k



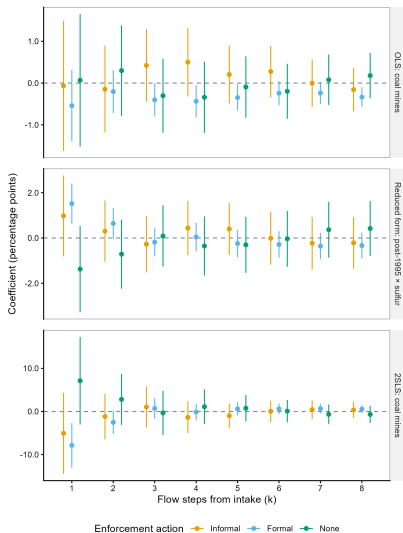
For utilities upstream of a mine — where mining cannot reach the intake hydrologically — OLS estimates on MR violations are positive and significant at every k . In the reduced form and 2SLS, nitrate estimates are significant for $k \geq 3$, while arsenic and inorganic chemical estimates are significant only at $k = 1$.

Appendix: exclusion placebo, visits across k



For utilities upstream of a mine, inspection visits rise significantly at nearly every k , sanitary visits at $k = 1$ and $k \geq 4$, and sample collection visits at several radii; technical assistance and enforcement visits are mostly insignificant.

Appendix: exclusion placebo, enforcement across k



Enforcement actions show no consistent pattern for utilities upstream of a mine at any choice of k , supporting the exclusion restriction.

References I